

TUSKIRA

Sentinel Alert Forwarder

Deployment Guide & Customer Setup

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Customer Confidential

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1 Overview

The Tuskira Sentinel Alert Forwarder is a Terraform module that automatically forwards Microsoft Sentinel alerts to the Tuskira platform. When Sentinel creates an incident from an analytic rule, an Azure Logic App extracts the alert details and sends them to your Tuskira API endpoint. All credentials are stored securely in Azure Key Vault.

Architecture

```
Architecture
Microsoft Sentinel
|
| (incident created by analytic rule)
v
Automation Rule
|
| (triggers playbook)
v
Azure Logic App
|
|--- Reads API credentials from Azure Key Vault
|--- Extracts alert details from the incident
|--- Base64-encodes each alert
|
v (HTTPS POST)
Tuskira API (/api/v2/alerts)
```

Resources Deployed

Resource	Name Pattern	Purpose
User-Assigned Managed Identity	tsk-{tenant}-{env}-alert-fwd-id	Authenticates Logic App to Sentinel and Key Vault
Logic App (Consumption)	tsk-{tenant}-{env}-sentinel-alert-fwd	Processes incidents and forwards alerts to Tuskira
Key Vault	{tenant}-{env}-fwd-kv	Stores Tuskira API URL and API key
Sentinel API Connection	tsk-{tenant}-{env}-sentinel-conn	Connects Logic App to Microsoft Sentinel
Key Vault API Connection	tsk-{tenant}-{env}-keyvault-conn	Connects Logic App to Key Vault
Automation Rules (optional)	Forward to Tuskira - {rule}	Triggers the Logic App for specific analytic rules

2 Prerequisites

Before deploying, ensure you have the following:

Azure Resources (must already exist)

- **Azure Resource Group** — A resource group where the forwarder resources will be deployed
- **Log Analytics Workspace** — With Microsoft Sentinel enabled
- **Microsoft Sentinel** — Enabled on the workspace with at least one active analytic rule

Tuskira Credentials

Contact your Tuskira representative to obtain:

- **API URL** — Your Tuskira API endpoint (e.g., <https://your-company.tuskira.ai/api/v2/alerts>)
- **API Key** — A Bearer token for authenticating API requests

Required Tools

Tool	Minimum Version	Purpose
Terraform	1.5.0	Infrastructure deployment
Azure CLI	2.50+	Azure authentication
Git	Any	Cloning the module repository

Azure CLI Authentication

```
bash
az login
az account set --subscription "YOUR_SUBSCRIPTION_ID"
```

3 Required Azure Permissions

The user or service principal running terraform apply must have the following permissions.

Recommended: Built-in Role Assignments

Role	Scope	Why
Contributor	Resource Group	Creates Logic App, Key Vault, managed identity, API connections, secrets
User Access Administrator	Resource Group	Creates role assignments for the managed identity and Azure Security Insights SP
Microsoft Sentinel Contributor	Log Analytics Workspace	Creates automation rules (only if using Terraform-managed rules)
Directory Readers	Azure AD Tenant	Reads the Azure Security Insights service principal

Assigning Resource Group Roles

```
bash
DEPLOYER_ID="<your-user-or-sp-object-id>"
RG_SCOPE="/subscriptions/<sub-id>/resourceGroups/<rg-name>"

az role assignment create --assignee $DEPLOYER_ID \
  --role "Contributor" --scope $RG_SCOPE

az role assignment create --assignee $DEPLOYER_ID \
  --role "User Access Administrator" --scope $RG_SCOPE
```

Sentinel Contributor on Workspace

```
bash
WORKSPACE_SCOPE="/subscriptions/<sub-id>/resourceGroups/<rg-name>/providers/
Microsoft.OperationalInsights/workspaces/<workspace-name>"

az role assignment create --assignee $DEPLOYER_ID \
  --role "Microsoft Sentinel Contributor" --scope $WORKSPACE_SCOPE
```

Azure AD Directory Readers

```
bash
az rest --method POST \
  --url "https://graph.microsoft.com/v1.0/directoryRoles/
roleTemplateId=88d8e3e3-8f55-4a1e-953a-9b9898b8876b/members/$ref" \
  --body "{\"@odata.id\":
  \"https://graph.microsoft.com/v1.0/directoryObjects/$DEPLOYER_ID\"}"
```

Detailed Permission Breakdown

Azure Resource	Permission	Scope
Resource Group	Microsoft.Resources/subscriptions/resourceGroups/read	Resource Group

Azure Resource	Permission	Scope
Managed Identity	Microsoft.ManagedIdentity/userAssignedIdentities/*	Resource Group
Logic App	Microsoft.Logic/workflows/*	Resource Group
API Connections	Microsoft.Web/connections/*	Resource Group
Key Vault	Microsoft.KeyVault/vaults/*	Resource Group
Key Vault Secrets	Microsoft.KeyVault/vaults/secrets/*	Key Vault
Role Assignments	Microsoft.Authorization/roleAssignments/write	RG + Workspace
Managed APIs	Microsoft.Web/locations/managedApis/read	Subscription
Sentinel Rules	Microsoft.SecurityInsights/automationRules/*	Workspace (if TF rules)
Azure AD	microsoft.directory/servicePrincipals/read	Tenant

4 Deployment Steps

Step 1 — Create Your Deployment Directory

```
bash
mkdir sentinel-forwarder-deploy
cd sentinel-forwarder-deploy
```

Step 2 — Create main.tf

```
hcl
terraform {
  required_version = ">= 1.5.0"

  required_providers {
    azurerm = { source = "hashicorp/azurerm", version = "~> 3.100" }
    azapi = { source = "Azure/azapi", version = "~> 1.12" }
    azuread = { source = "hashicorp/azuread", version = "~> 2.47" }
  }
}

provider "azurerm" {
  features {
    key_vault { purge_soft_delete_on_destroy = false }
  }
}
provider "azapi" {}
provider "azuread" {}

variable "api_url" {
  description = "Tuskira API endpoint URL"
  type = string
}

variable "api_key" {
  description = "Tuskira API Bearer token"
  type = string
  sensitive = true
}

module "sentinel_forwarder" {
  source = "git::ssh://git@bitbucket.org/deepinsight-team/tuskira-sentinel-alert-forwarder.git?ref=main"

  tenant_name = "your-company"
  environment = "prod"
  resource_group_name = "your-resource-group"
  location = "australiaeast"
  log_analytics_workspace_id = "/subscriptions/SUB_ID/resourceGroups/RG/providers/Microsoft.OperationalInsights/workspaces/WORKSPACE"

  api_url = var.api_url
  api_key = var.api_key

  # enable_automation_rules = true
  # alert_rule_ids = [ ... ]
}

output "logic_app_name" {
  value = module.sentinel_forwarder.logic_app_name
}
output "logic_app_id" {
  value = module.sentinel_forwarder.logic_app_id
}
```

Replace these placeholder values:

Placeholder	Where to Find It
your-company	A short identifier for your organization
prod	Your environment name (dev, staging, prod)
your-resource-group	Azure Portal > Resource Groups
australiaeast	Your preferred Azure region
log_analytics_workspace_id	Azure Portal > Log Analytics Workspaces > Properties > Resource ID

Step 3 — Create terraform.tfvars

Create a file named `terraform.tfvars` with your Tuskira credentials. **Do not commit this file to version control.**

```
hcl
api_url = "https://your-company.tuskira.ai/api/v2/alerts"
api_key = "your-tuskira-api-key"
```

Step 4 — Initialize and Deploy

```
bash
terraform init
terraform plan # Review what will be created
terraform apply # Deploy (type 'yes' to confirm)
```

Expected output after successful deployment:

```
output
Apply complete! Resources: 16 added, 0 changed, 0 destroyed.

Outputs:

logic_app_name = "tsk-your-company-prod-sentinel-alert-fwd"
logic_app_id = "/subscriptions/.../workflows/tsk-your-company-prod-sentinel-alert-fwd"
```

Step 5 — Connect Analytic Rules to the Forwarder

Option A: Sentinel Portal UI (Recommended for Getting Started)

1. Open **Azure Portal** > **Microsoft Sentinel** > your workspace
2. Go to **Automation** in the left menu
3. Click **Create** > **Automation rule**
4. Configure: Name (e.g., "Forward alerts to Tuskira"), Trigger: "When incident is created", Conditions (optional): Filter by analytic rule name, severity, etc., Actions: Select **Run playbook**, then choose the Logic App name from the output
5. Click **Apply**

Option B: Terraform-Managed Automation Rules

Uncomment the automation rule variables in your `main.tf`:

```
hcl
enable_automation_rules = true
alert_rule_ids = [
  "/subscriptions/SUB_ID/resourceGroups/RG/providers/
  Microsoft.OperationalInsights/workspaces/WS/providers/
  Microsoft.SecurityInsights/alertRules/RULE_NAME",
]
```

To find your analytic rule IDs:

```
bash
az sentinel alert-rule list \
--workspace-name YOUR_WORKSPACE \
--resource-group YOUR_RG \
--query "[].{name:name, displayName:displayName}" \
-o table
```

5 Verification

Check the Logic App

- **Azure Portal** > **Resource Groups** > your resource group
- Find the Logic App (name starts with tsk-)
- Verify **Status** is "Enabled"
- Check the **Trigger** shows "Microsoft_Sentinel_incident"

Check Automation Rules

- **Azure Portal** > **Microsoft Sentinel** > **Automation**
- Verify your automation rule is listed and enabled
- The rule should show the Logic App as the playbook action

Test Alert Forwarding

Trigger a test alert in Sentinel to verify the full pipeline:

1. Create or wait for a Sentinel analytic rule to fire
2. An incident will be created in Sentinel
3. The automation rule triggers the Logic App
4. Check the Logic App **Run history** in the Azure Portal
5. A successful run shows status "Succeeded" with a 200 response from the Tuskira API

Check Logic App Run History

- Open the Logic App in Azure Portal
- Click **Run history** in the Overview pane
- Each run shows: **Status** (Succeeded / Failed), **Trigger** (Microsoft_Sentinel_incident), **Duration** (typically 2–5 seconds)
- Click on a run to see the detailed execution of each step

6 Updating Credentials

If you need to rotate your Tuskira API key:

1. Update the `api_key` value in your `terraform.tfvars`
2. Run `terraform apply`
3. The new key is stored in Key Vault and used by the Logic App on the next run

No restart or redeployment of the Logic App is needed.

7 Troubleshooting

"Missing required permissions for Microsoft Sentinel on the playbook"

Cause: The Azure Security Insights service principal does not have the Sentinel Automation Contributor role on the Logic App.

Fix: Run `terraform apply` again. The module automatically creates this role assignment. If it persists, verify your deployer has User Access Administrator on the resource group.

"Forbidden" errors in Logic App run history

Cause: The Logic App's managed identity does not have the correct roles.

Fix: Verify the Microsoft Sentinel Responder role exists on the workspace:

```
bash
az role assignment list \
--scope "/subscriptions/SUB_ID/resourceGroups/RG/providers/
Microsoft.OperationalInsights/workspaces/WORKSPACE" \
--query "[?roleDefinitionName=='Microsoft Sentinel Responder']" \
-o table
```

Logic App runs are not being triggered

Cause: No automation rule is connecting the analytic rule to the Logic App.

Fix: - If using Portal UI: Check Sentinel > Automation for your rule - If using Terraform: Verify `enable_automation_rules = true` and `alert_rule_ids` contains the correct rule IDs - Verify the analytic rule is enabled and creating incidents

Key Vault access denied

Cause: The Logic App's managed identity lost its Key Vault access policy.

Fix: Run `terraform apply` to recreate the access policy.

"Directory.Read.All" or service principal lookup fails

Cause: The deployer does not have Azure AD read permissions.

Fix: Ask your Azure AD administrator to assign the **Directory Readers** role:

```
bash
az ad signed-in-user show --query id -o tsv
# Give this object ID to your admin to add to Directory Readers role
```

8 Uninstalling

To remove all deployed resources:

```
bash
terraform destroy
```

This removes the Logic App, Key Vault, managed identity, connections, role assignments, and automation rules. It does **not** affect your Sentinel workspace, analytic rules, or existing incidents.

Support

For questions about this deployment, contact your Tuskira representative.